

Is MARINA Under-predicting the Rare Bits?

Per-bit calibration of the 16,384-bit RankingEntropy fingerprint by bit index

Analysis over Checkpoints/MARINA/final1, MARINA1 test split

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Abstract

The 16,384 bits of MARINA’s **RankingEntropy** (“sherlock”) fingerprint are ordered by descending marginal entropy, which for this bit pool is exactly descending frequency: bit 0 fires in 3.85% of test molecules, bit 16,383 in 0.04%. A natural worry is that the multi-label head learns the frequent head of that distribution and hedges on the rare tail, systematically under-predicting late bits. **It does not.** The model under-predicts by roughly 3% of total mass, but that deficit is *uniform in bit index*, not progressive. Micro-F1 steps from 0.979 over the first 1024 bits to ≈ 0.95 and then stays in 0.936–0.950 for the remaining 15,360 bits while frequency falls 96 $\times$; per-bit AUC holds at 0.993–0.998 throughout; and reliability diagrams for common and rarest bits are superimposable. Decomposing F1 shows that precision is flat by index and *all* index variation lives in recall — the model prefers to miss a bit rather than invent one. Stratifying by whether a bit is logically implied by another bit (10,608 of 16,384 are) confirms the flatness is not an artifact of redundant bits carrying the tail: both strata are separately flat, and both trend mildly *upward* toward rarer bits.

1 Question and setup

`EntropyFPLoader.setup (src/modules/data/fp_loader.py)` ranks candidate circular substructures by marginal binary entropy and keeps the top 16,384:

```
topk_sorted = sorted(topk_idx, key=lambda i: (-ent[i], bitinfos[i]))
```

Binary entropy is monotone in p below 0.5 and every selected bit has $p \ll 0.5$, so this ordering is equivalent to descending frequency (Figure 1a). *Later bit index = rarer substructure* is therefore the axis of interest, and the question is whether the predicted probability q_j for bit j falls short of its true rate p_j more severely as j grows.

Data. Cached sigmoid outputs for $n = 3000$ MARINA1 test molecules against the checkpoint `Checkpoints/MARINA/final1` (`results/preds.npz`, produced by `01_predict.py`). These are the same predictions that reproduce the previously reported top-1 retrieval of 0.7587, so the analysis inherits a known-good model state. Ground truth averages 58.3 set bits per molecule against 57.5 units of predicted mass; mean cosine to truth is 0.960 (median 0.991).

Redundancy stratification. From `02_bit_mi.py`, $\max_{j \neq i} \text{cond}[i] = \max_{j \neq i} P(\text{bit } i = 1 \mid \text{bit } j = 1)$, computed over the **518,901-molecule rankingset** — an independent corpus, not the test molecules. Bit i is **implied** when this equals 1, i.e. $\text{supp}(j) \subseteq \text{supp}(i)$ for some other bit j ; **independent** otherwise. 10,608 of 16,384 bits are implied, and they concentrate heavily at low indices (Figure 1b), so any unstratified score-vs-index curve mixes two effects and must be split.

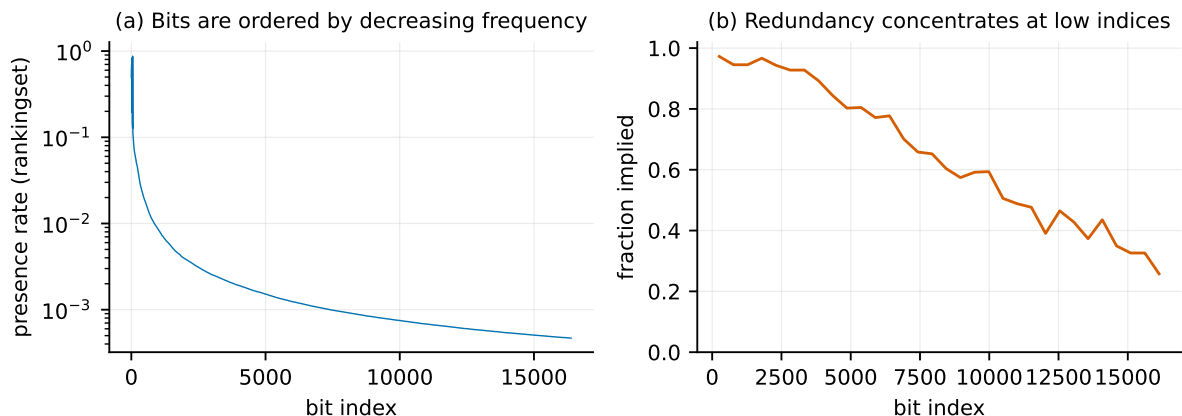


Figure 1: The bit axis. (a) Presence rate over the 518,901-molecule rankingset falls monotonically with bit index, confirming that the entropy criterion is a frequency criterion. (b) The fraction of bits that are logically implied by some other bit falls from $\approx 96\%$ at low index to $\approx 32\%$ at high index — the reason every index-dependent claim below is reported within redundancy strata.

2 Result 1: mass bias is flat, not progressive

For each of 16 equal bins define the mass ratio $\sum_j q_j / \sum_j p_j$, bootstrapped over molecules ($B = 2000$). If the model hedged on rare bits this ratio would decay with index. It does not (Figure 2, Table 1): there is a one-time step from 0.995 over the first 1024 bits to ≈ 0.97 , after which the ratio is flat to bit 16,383 — the final bin (0.9760) sits *above* eight of the middle bins. Per molecule the shortfall is 57.5 predicted mass against 58.3 true bits.

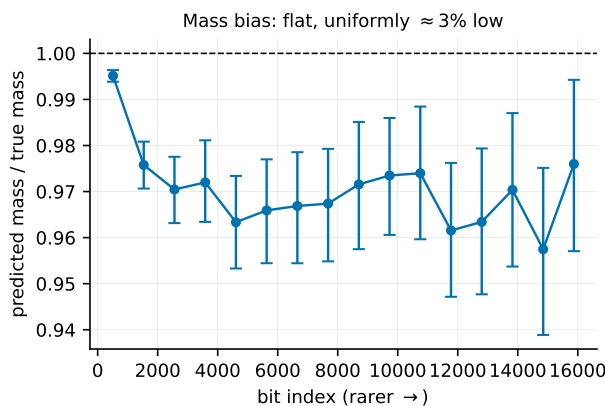


Figure 2: Predicted mass over true mass by bit index, with 95% bootstrap CIs over molecules. A uniform $\approx 3\%$ deficit, not a decay. Widening CIs at high index reflect positive counts thinning out, not increasing bias.

A per-bit Spearman correlation between relative bias $(q_j - p_j)/p_j$ and index is weakly negative and significant ($\rho = -0.125$, $p = 8.7 \times 10^{-19}$, over the 5007 bits with ≥ 5 positives). This is worth stating precisely rather than either suppressing or over-reading: the *median* per-bit bias is -0.0006 while the *mean* is -0.031 . The bulk of late bits are unbiased and the correlation is carried by a minority tail of badly-missed bits, which is a different phenomenon from a shift of the distribution.

Table 1: Per-bin calibration and micro-averaged classification at threshold 0.5. **Precision varies by 1.2 points across bins 2–16 while recall varies by 2.1 and drives every F1 movement.** AUC is the mean over bits of the within-bit across-molecule ranking AUC.

bit range	true rate	q/p	95% CI	prec.	recall	F1	FN	FP	AUC
0–1023	0.03849	0.9951	[0.9938, 0.9964]	0.9819	0.9753	0.9786	2915	2129	0.9982
1024–2047	0.00499	0.9758	[0.9707, 0.9808]	0.9651	0.9360	0.9503	980	518	0.9975
2048–3071	0.00268	0.9705	[0.9632, 0.9775]	0.9664	0.9347	0.9503	538	268	0.9976
3072–4095	0.00191	0.9720	[0.9634, 0.9811]	0.9595	0.9282	0.9436	422	230	0.9967
4096–5119	0.00145	0.9633	[0.9533, 0.9734]	0.9614	0.9215	0.9410	350	165	0.9946
5120–6143	0.00120	0.9659	[0.9544, 0.9770]	0.9615	0.9243	0.9425	278	136	0.9956
6144–7167	0.00101	0.9669	[0.9544, 0.9786]	0.9594	0.9214	0.9400	244	121	0.9962
7168–8191	0.00081	0.9674	[0.9549, 0.9793]	0.9653	0.9299	0.9472	174	83	0.9950
8192–9215	0.00072	0.9716	[0.9575, 0.9851]	0.9548	0.9214	0.9378	175	97	0.9950
9216–10239	0.00066	0.9735	[0.9606, 0.9860]	0.9615	0.9358	0.9485	130	76	0.9959
10240–11263	0.00058	0.9740	[0.9596, 0.9884]	0.9549	0.9280	0.9412	128	78	0.9969
11264–12287	0.00055	0.9615	[0.9472, 0.9762]	0.9592	0.9151	0.9367	144	66	0.9959
12288–13311	0.00053	0.9634	[0.9477, 0.9794]	0.9574	0.9188	0.9377	131	66	0.9945
13312–14335	0.00046	0.9704	[0.9537, 0.9870]	0.9613	0.9255	0.9431	106	53	0.9967
14336–15359	0.00046	0.9575	[0.9389, 0.9751]	0.9589	0.9145	0.9362	122	56	0.9930
15360–16383	0.00040	0.9760	[0.9571, 0.9943]	0.9570	0.9295	0.9430	86	51	0.9974

3 Result 2: precision is flat, recall carries everything

Micro-F1 steps from 0.9786 to ≈ 0.95 after the first 1024 bits and then holds between 0.9362 and 0.9503 across the remaining 15,360 bits, over which the base rate falls by a factor of 96. The last bin (0.9430) beats four middle bins. Per-bit AUC is 0.993–0.998 with no trend. Macro (per-bit) F1 is deliberately not reported: in the tail most bits have ≈ 1 positive across 3000 molecules, making per-bit F1 zero-or-one noise.

The decomposition is the informative part. **Precision is essentially constant** at 0.955–0.966 across bins 2–16, so the entire index dependence of F1 is a recall effect (Figure 3a). Precision exceeds recall in every single bin, equivalently $FN > FP$ in every bin (Figure 4b) — the same $\approx 3\%$ under-prediction seen in the mass ratio, now visible as an error asymmetry. The model would rather miss a substructure than hallucinate one, which for downstream retrieval is the benign direction.

4 Result 3: the flatness survives the redundancy control

Because implied bits fall from 96% to 32% of each octile, and implied bits score higher, an unstratified curve could hide a genuine decay. It does not (Table 2): within each stratum the trend across the tail is flat or mildly *increasing* — independent bits go $0.9018 \rightarrow 0.9201$ in F1 from octile 2 to octile 8, implied bits $0.9515 \rightarrow 0.9815$. The persistent ≈ 6 -point gap between strata is a recall gap (precision differs by ≈ 4 points, recall by ≈ 8), i.e. independent bits are *missed* more often rather than false-alarmed more often.

What “implied” means chemically. The implication runs *specific* $\rightarrow$ *general*, which is the opposite of what the name suggests. Over a 300-bit sample of implied bits, the guaranteeing bit j is a *later*, *rarer* bit 83.7% of the time (median rankingset count 427 versus 595 for the bit it implies) and a *larger-radius* fragment 66.7% of the time. This is ECFP radius nesting: a large substructure contains the small core inside it, so the small core is forced on whenever the large one fires (Table 3). An implied bit is therefore easy not because a common bit gives it

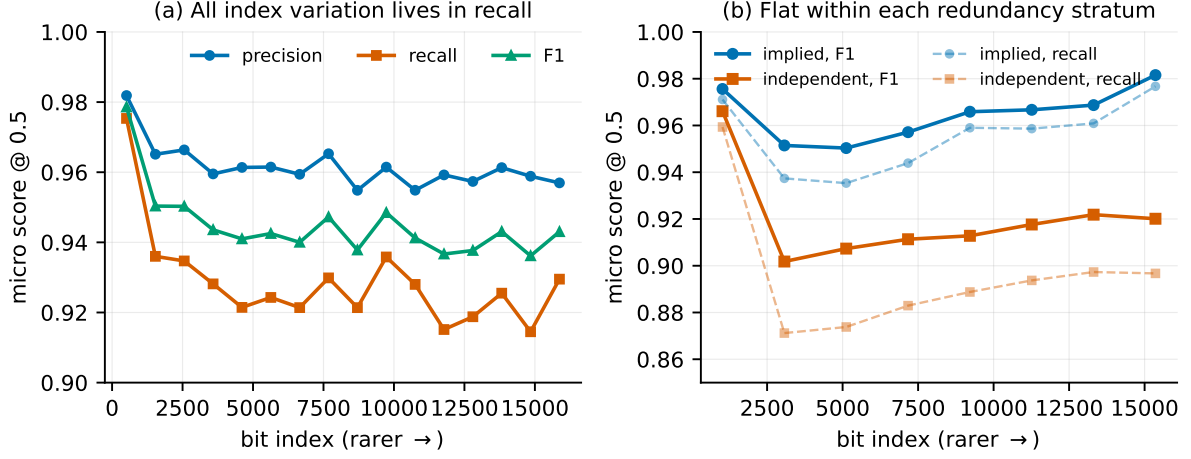


Figure 3: Micro precision / recall / F1 at threshold 0.5. (a) Precision is flat by index; recall carries all the variation, and the gap between them is the under-prediction. (b) Split by redundancy stratum: both implied and independent bits are flat-to-rising toward rarer bits, and the ≈ 6 -point gap between strata is itself a recall gap, stable at every index.

Table 2: Micro P/R/F1 at 0.5, stratified by whether a bit is logically implied by another bit. Both strata are separately flat-to-rising with index, so the flat aggregate is not an artifact of redundant bits propping up the tail.

bit range	% impl.	implied			independent			n_+ (indep.)
		prec.	recall	F1	prec.	recall	F1	
0–2047	95.8%	0.9802	0.9711	0.9756	0.9730	0.9593	0.9661	3075
2048–4095	92.3%	0.9660	0.9374	0.9515	0.9346	0.8712	0.9018	1149
4096–6143	80.6%	0.9658	0.9353	0.9503	0.9435	0.8738	0.9073	1664
6144–8191	69.7%	0.9708	0.9439	0.9571	0.9416	0.8829	0.9113	1717
8192–10239	59.1%	0.9728	0.9590	0.9659	0.9382	0.8888	0.9128	1861
10240–12287	46.5%	0.9749	0.9586	0.9667	0.9429	0.8937	0.9176	1976
12288–14335	42.5%	0.9767	0.9609	0.9687	0.9478	0.8973	0.9218	1860
14336–16383	31.5%	0.9864	0.9767	0.9815	0.9447	0.8967	0.9201	1830

away, but because it is a coarse feature with many possible witnesses — detecting *any* enclosing fragment forces it. Pairs with identical counts are the 1,695 two-way cases, where containment is mutual and the columns are exact duplicates.

Table 3: Sampled implied bits and the bit that guarantees them (r = ECFP radius, c = rankingset count). The antecedent is typically the rarer, larger fragment that structurally contains the consequent.

bit i	fragment i (implied)	r_i	c_i	bit j	fragment j (antecedent)	r_j	c_j
4608	CC(C)NC(N)=O	2	854	9964	CCC1NC(=O)N(C)C1=O	3	390
6983	CCC(OC)(OC)C(=O)O	2	551	7385	COC1(C(=O)O)CCC(N)C(C(C)O)O1	3	520
10249	CCC(C(C)(C)C)C(C)(C)O	2	379	10392	CCC1C(C)(O)CC=C2CCCCC21C	3	374
10937	CNc1ccc(C)cc1	3	355	12108	CNc1ccc(C(OC)C(C)N)cc1	4	322
11081	cc(C)c1cc2c(C)coc2cc1o	4	351	11085	Cc1coc2cc3oc(=O)cc(C)c3cc12	4	351
11719	CNC(=O)C(O)C(C)(C)COP(=O)(O)OP(=O)(O)O	5	333	11701	CC(C)(COP(=O)(O)OP)C(O)C(N)=O	4	333

5 Result 4: over-confidence is index-independent

Whether rare bits are miscalibrated *as probabilities* is a separate question from whether they are ranked or thresholded well. Binning predictions by confidence and measuring the empirical rate (Table 4, Figure 4a) gives four essentially superimposable curves. The model is over-confident by the same margin everywhere: predictions near 0.964 are correct 87%, 88%, 85%, 85% of the time for common, mid, rarest, and rarest-independent bits respectively. There is no rare-bit-specific miscalibration to correct.

The one quantity that does move is the miss rate: 2.5% of positives fall below 0.5 for the common bits, 6.9% mid, 7.6% rarest, and 10.6% for rarest-and-independent — again a recall effect, and again a step rather than a slope.

Table 4: Reliability by confidence bucket. Rows are near-identical across bit groups; the model’s over-confidence does not depend on bit rarity.

	0–1023 (common)		8192–16383 (rarest)		8192–16383 indep.	
confidence	<i>n</i>	empirical	<i>n</i>	empirical	<i>n</i>	empirical
[0.5, 0.7)	1081	0.5273	302	0.5397	237	0.5316
[0.7, 0.9)	1783	0.6826	508	0.6831	380	0.7263
[0.9, 0.99)	4543	0.8734	1096	0.8522	844	0.8531
[0.99, 0.999)	8060	0.9695	1602	0.9713	1129	0.9681
[0.999, 1)	101977	0.9977	9423	0.9963	4544	0.9936
missed positives	0.0247		0.0762		0.1059	

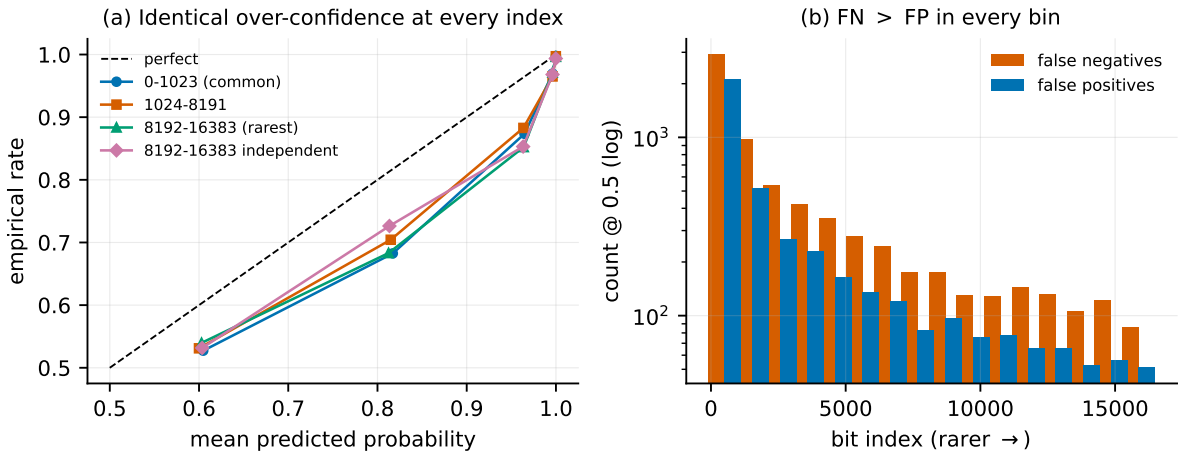


Figure 4: (a) **Reliability diagrams** for four bit groups spanning the full rarity range; the curves lie on top of each other, all below the diagonal by the same margin. (b) **False negatives exceed false positives in every bin** (log scale), the error-side view of the uniform $\approx 3\%$ mass deficit.

6 Caveats

1. **Tail sample size.** 3000 test molecules means the final bin rests on 1,220 positives and only 42 of its 2048 bits have ≥ 5 positives. The ± 0.01 wobble between adjacent late bins is not interpretable; the flatness claim rests on the absence of a trend across sixteen bins, not on any single bin. Re-running `01_predict.py --limit 0` over the full test split would tighten this.

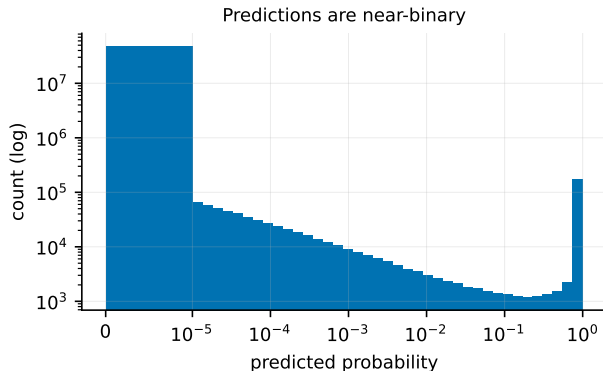


Figure 5: Predictions are near-binary. 99.26% of the 49.2M predicted probabilities fall below 10^{-4} and 0.34% exceed 0.9, leaving $<0.06\%$ in the middle. The head is decisive rather than hedging — consistent with a model that is not shrinking rare-bit outputs toward the base rate.

2. **Threshold choice.** P/R/F1 are reported at 0.5. Given the near-binary output distribution (Figure 5) the numbers are insensitive to this within a wide range, but they are not threshold-free; AUC and the mass ratio are, and they agree.
3. **Single checkpoint, single split.** All results are `final1` on the MARINA1 test split. Nothing here is seed-averaged.
4. **The Spearman trend is real but small.** Reported in Section 2 rather than dismissed: $\rho = -0.125$ is statistically solid and directionally consistent with mild extra under-prediction of rare bits. The claim being defended is that its magnitude (median bias -0.0006) is negligible relative to the $96\times$ change in base rate, not that it is zero.

7 Conclusion

MARINA is learning the rare tail of the fingerprint appropriately. There is a genuine under-prediction bias — about 3% of mass, visible as $\text{FN} > \text{FP}$ in every bin and as precision-above-recall throughout — but it is uniform in bit index rather than concentrated in the rare bits. Across the 15,360 bits after the first 1024, base rate falls $96\times$ while F1 stays within 0.936–0.950, AUC within 0.993–0.998, and reliability curves remain superimposable. The single index-dependent effect in the entire analysis is recall, and it is a one-time step at bit ≈ 1024 , not a decay. Stratifying by logical implication — necessary because redundant bits are $3\times$ over-represented at low indices — leaves both strata separately flat.

The practical implications: (i) there is no case for a rarity-dependent loss reweighting or per-bit threshold calibration, since neither the bias nor the over-confidence depends on index; (ii) the uniform under-prediction is in the safe direction for retrieval and could be removed globally with a single threshold shift if desired; (iii) the wasted capacity documented in the companion MI-redundancy analysis is a *selection* problem, not a *learning* problem — the model extracts what is there, so any gain must come from choosing better bits rather than from fitting the current ones harder.

Reproduce: `cd Datasets/fp-redundancy && pixi run python scripts/10_bit_calibration.py (tables), scripts/11_report_figures.py (figures + report/stats.json).` Both consume `results/preds.npz` and `results/bit_mi.npz`. **Compile:** `tectonic report/bit-calibration.tex`.